Goal

The representation file constructs a homomorphism

$$\rho : \mathrm{SolutionGroup}(S) \to (n \times n\ \ \mathbb{C})^{\times}$$

from matrices satisfying the analytic identities extracted by the EPR argument.

The generator assignment is

$$\rho_0(x_j) = \mathrm{obs}_j, \qquad \rho_0(J) = -I.$$

Since the solution group is a presented group, this generator assignment gives a homomorphism only after every defining relator maps to 1:

$$r \in \mathrm{Rel}(S) \Longrightarrow \mathrm{lift}(\rho_0)(r) = 1.$$

The file is mainly about organizing these relator proofs.

Matrix units

The target is the group of units

$$(n \times n\ \ \mathbb{C})^{\times}.$$

An observable matrix satisfies

$$\mathrm{obs}_j^2 = I.$$

So it is packaged as a unit by taking itself as its own inverse:

$$\widehat{\mathrm{obs}_j} = (\mathrm{obs}_j, \mathrm{obs}_j), \qquad \widehat{\mathrm{obs}_j}^{\,2} = 1.$$

The distinguished generator J is represented by

$$-I.$$

Since

$$(-I)^2 = I,$$

it also gives a unit:

$$\widehat{-I}^2 = 1.$$

Thus the generator-level map used in Lean is

$$\rho_0(x_j) = \widehat{\mathrm{obs}_j}, \qquad \rho_0(J) = \widehat{-I}.$$

This is `solutionGroupGeneratorImage`.

The relators

The solution group has generators

$$x_j \qquad \text{for variables } j, \qquad \text{and} \qquad J.$$

There are five kinds of relators.

First, every variable is an involution:

$$x_j^2 = 1.$$

Second, J is an involution:

$$J^2 = 1.$$

Third, every variable commutes with J:

$$x_j J x_j^{-1} J^{-1} = 1,$$

equivalently

$$x_j J = J x_j.$$

Fourth, variables that occur in a common equation commute:

$$x_j x_k x_j^{-1} x_k^{-1} = 1,$$

equivalently

$$x_j x_k = x_k x_j.$$

Fifth, each equation gives a row-product relator. If V_i is the support of row i, then

$$\prod_{j \in V_i} x_j = J^{b_i}.$$

As a relator word this is written

$$\left(\prod_{j \in V_i} x_j \right) J^{-b_i} = 1.$$

In Lean this is `equationRelator S i`.

What must be proved

To construct the homomorphism, we must prove that each of these relator families is killed by ρ_0.

For variable involutions,

$$\mathrm{lift}(\rho_0)\left(x_j^2\right) = \rho_0\left(x_j\right)^2 = \widehat{\mathrm{obs}_j}^2 = 1.$$

This uses observability:

$$\mathrm{obs}_j^2 = I.$$

For the J involution,

$$\mathrm{lift}(\rho_0)\left(J^2\right) = \rho_0(J)^2 = \widehat{-I}^2 = 1.$$

This is purely algebraic.

For centrality of J,

$$\mathrm{lift}(\rho_0)\left(x_j J x_j^{-1} J^{-1}\right) = 1$$

is the same as

$$\rho_0\left(x_j\right)\rho_0(J) = \rho_0(J)\rho_0\left(x_j\right).$$

Since $\rho_0(J) = -I$, this holds because scalar matrices commute with every matrix:

$$\mathrm{obs}_j(-I) = (-I)\,\mathrm{obs}_j\,.$$

This is `commute_negOneMatrixUnit`.

For same-equation commutation, assume j and k occur together in a row. The relator asks for

$$\rho_0(x_j)\rho_0(x_k) = \rho_0(x_k)\rho_0(x_j).$$

At the matrix level this is

$$\mathrm{obs}_j\,\mathrm{obs}_k = \mathrm{obs}_k\,\mathrm{obs}_j\,.$$

The observable strategy already carries row-wise commutation, and the Lean proof translates common-row membership into `sameEquation`.

This is handled by `sameEquation_comm_of_row_comm`.

The only nontrivial family left is the equation relator:

$$\mathrm{lift}(\rho_0)\left(\left(\prod_{j\in V_i} x_j\right)J^{-b_i}\right) = 1.$$

This is equivalent to proving the row identity

$$\prod_{j\in V_i} \mathrm{obs}_j = (-1)^{b_i} I.$$

The rest of the file is mainly a bridge from the EPR extraction theorem to this equation-relator proof.

Universal property step

Let R_S be the set of all relators. If

$$\forall r \in R_S, \qquad \mathrm{lift}(\rho_0)(r) = 1,$$

then the universal property of the presented group gives

$$\rho : \mathrm{SolutionGroup}(S) \to (n \times n\ \ \mathbb{C})^{\times}.$$

This is `solutionGroupRepresentationOfRelatorProof`.

The lemma `solutionGroupRelatorProofOfEquationProof` splits the proof by cases on the five relator families:

$$x_j^2, \qquad J^2, \qquad [x_j, J], \qquad [x_j, x_k], \qquad \left(\prod_{j\in V_i} x_j\right)J^{-b_i}.$$

The first two are solved automatically by the unit packaging. The J-commutation relators use $-I$. The same-equation commutation relators use the row-wise commutation hypothesis. The equation relators are supplied separately.

So the practical constructor asks only for:

$$\mathrm{obs}_j\,\mathrm{obs}_k = \mathrm{obs}_k\,\mathrm{obs}_j \qquad \text{when} \qquad j,k \qquad \text{share an equation},$$

and

$$\mathrm{lift}(\rho_0)(\mathrm{equationRelator}_i) = 1 \qquad \text{for every} \qquad i.$$

This is `solutionGroupRepresentationOfEquationProof`.

Equation words and row products

The equation word is the ordered free-group product

$$w_i = \prod_{j \in V_i} x_j.$$

The matrix-side row product is

$$\mathrm{Row}_{i(\mathrm{obs})} = \prod_{j \in V_i} \mathrm{obs}_j .$$

Lean fixes a concrete order using the sorted support list. This matters because free-group words are ordered, while the mathematical row product is order-independent only after proving the row observables commute.

The evaluator lemma says

$$\mathrm{val}(\mathrm{lift}(\rho_0)(w_i)) = \mathrm{Row}_{i(\mathrm{obs})} .$$

This is `lift_equationWord_toLinearSystem_val`.

Therefore, if

$$\mathrm{Row}_{i(\mathrm{obs})} = (-1)^{b_i} I,$$

then

$$\mathrm{lift}(\rho_0)(w_i) = \rho_0(J)^{b_i}.$$

Since

$$\rho_0(J) = -I,$$

the right hand side is exactly

$$(-I)^{b_i}.$$

Thus the relator

$$w_i J^{-b_i}$$

maps to

$$\rho_0(J)^{b_i} \rho_0(J)^{-b_i} = 1.$$

This is `lift_equationRelator_toLinearSystem_of_row`.

Where EPR enters

The EPR extraction theorem gives, for each row i,

$$\mathrm{Row}_{i(\mathrm{obs})} = (-1)^{b_i} I.$$

The representation file obtains this from local-loss annihilation. Choose some $j \in V_i$. The local loss at (i, j) gives the three identities from the EPR note:

$$A = B^T, \qquad \mathrm{Row} = (-1)^{b_i} I, \qquad (-1)^{b_i}\, \mathrm{Row}\, AB^T = I.$$

For the representation construction, only the middle identity is needed:

$$\mathrm{Row} = (-1)^{b_i} I.$$

The Lean proof identifies the strategy row product with the Alice lift of the concrete row observable product:

$$\mathrm{Alice\_Row\_Prod} = \mathrm{Row}_{i(\mathrm{obs})} \otimes I.$$

Then `local_matrix_identities_of_local_loss_annihilate_epr` returns

$$\mathrm{Row}_{i(\mathrm{obs})} = (-1)^{b_i} I.$$

This is `rowObservableProduct_eq_sign_of_local_loss`.

Final constructor

The final construction is:

$$\text{local loss kills} \qquad |\Omega\rangle \Longrightarrow \mathrm{Row}_{i(\mathrm{obs})} = (-1)^{b_i} I$$

$$\mathrm{Row}_{i(\mathrm{obs})} = (-1)^{b_i} I \Longrightarrow \mathrm{lift}(\rho_0)(\mathrm{equationRelator}_i) = 1$$

$$\text{all relators map to} \qquad 1 \Longrightarrow \rho : \mathrm{SolutionGroup}(\mathrm{game.toLinearSystem}) \to (n \times n \ \ \mathbb{C})^{\times}.$$

The resulting homomorphism satisfies

$$\rho(x_j) = \widehat{\mathrm{obs}}_j, \qquad \rho(J) = \widehat{-I}.$$

This is `solutionGroupRepresentationOfEPRLoss`.